	$\mathcal{K}^{\#1}_{0^+}$	$\mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi}$
$\mathcal{K}^{\#1}_{0^+}{}^{\dagger}$	0	0
$\mathcal{K}^{\#1}_{0^-}{}^{\dagger\alpha\beta\chi}$	0	0

	$\mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta}$	$\mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta}$	$\mathcal{K}^{\#1}_{1^-}{}_{\alpha}$	$\mathcal{K}^{\#2}_{1^-}{}_{\alpha}$
$\mathcal{K}^{\#1}_{1^+}{}^{\dagger\alpha\beta}$	$-\frac{Y_1 k^2}{2}$	$\frac{k^2 \overset{(4)}{\kappa}_4}{2\sqrt{2}}$	0	0
$\mathcal{K}^{\#2}_{1^+}{}^{\dagger\alpha\beta}$	$\frac{k^2 \overset{(4)}{\kappa}_4}{2\sqrt{2}}$	$\frac{Y_2 k^2}{2}$	0	0
$\mathcal{K}^{\#1}_{1^-}{}^{\dagger\alpha}$	0	0	0	0
$\mathcal{K}^{\#2}_{1^-}{}^{\dagger\alpha}$	0	0	0	0

	$\mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta}$	$\mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi}$
$\mathcal{K}^{\#1}_{2^+}{}^{\dagger\alpha\beta}$	$\frac{3 k^2 \overset{(4)}{\kappa}_1}{2}$	0
$\mathcal{K}^{\#1}_{2^-}{}^{\dagger\alpha\beta\chi}$	0	0

	$\mathcal{J}^{\#1}_{0^+}$	$\mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi}$
$\mathcal{J}^{\#1}_{0^+}{}^{\dagger}$	0	0
$\mathcal{J}^{\#1}_{0^-}{}^{\dagger\alpha\beta\chi}$	0	0

	$\mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta}$	$\mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta}$	$\mathcal{J}^{\#1}_{1^-}{}_{\alpha}$	$\mathcal{J}^{\#2}_{1^-}{}_{\alpha}$
$\mathcal{J}^{\#1}_{1^+}{}^{\dagger\alpha\beta}$	$\frac{Y_2 k^2}{2\,\mathrm{Det}(1^+)}$	$-\frac{k^2 \overset{(4)}{\kappa}_4}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	0	0
$\mathcal{J}^{\#2}_{1^+}{}^{\dagger\alpha\beta}$	$-\frac{k^2 \overset{(4)}{\kappa}_4}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	$-\frac{Y_1 k^2}{2\,\mathrm{Det}(1^+)}$	0	0
$\mathcal{J}^{\#1}_{1^-}{}^{\dagger\alpha}$	0	0	0	0
$\mathcal{J}^{\#2}_{1^-}{}^{\dagger\alpha}$	0	0	0	0

	$\mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta}$	$\mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi}$
$\mathcal{J}^{\#1}_{2^+}{}^{\dagger\alpha\beta}$	$\frac{2}{3 k^2 \overset{(4)}{\kappa}_1}$	0
$\mathcal{J}^{\#1}_{2^-}{}^{\dagger\alpha\beta\chi}$	0	0

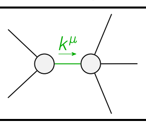
Abbreviations used in matrices

$$Y_1 == 3\overset{(4)}{\kappa}_1 + 2\overset{(4)}{\kappa}_3 \;\&\&\; Y_2 == \overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4 \;\&\&\; \mathrm{Det}(1^+) == -\frac{1}{8} k^4 \left(6\overset{(4)}{\kappa}_1 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) + (2\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2\right)$$

Lagrangian

$$\overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^{\delta}{}_{\chi\delta} \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^{\delta} + \overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^{\delta} -$$
$$3\overset{(4)}{\kappa}_1 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} - \overset{(4)}{\kappa}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} + \frac{1}{2}\overset{(4)}{\kappa}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{2}\overset{(4)}{\kappa}_4 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	**# constraint(s)**	**Covariant form**
$\mathcal{J}^{\#1}_{0^-}{}^{\alpha\beta\chi} == 0$	1	$\partial_\delta\partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta \mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta \mathcal{J}^{\delta\chi\alpha} + \partial_\delta\partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta \mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}^{\#1}_{0^+} == 0$	1	$\partial_\beta \mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$
$\mathcal{J}^{\#2}_{1^-}{}^{\alpha} == 0$	3	$\partial_\chi\partial_\beta \mathcal{J}^{\beta\alpha\chi} == 0$
$\mathcal{J}^{\#1}_{1^-}{}^{\alpha} == 0$	3	$\partial_\chi\partial^\alpha \mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi \mathcal{J}^{\beta\alpha}{}_{\beta} == \partial_\chi\partial_\beta \mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}^{\#1}_{2^-}{}^{\alpha\beta\chi} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha \mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon \mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta \mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon \mathcal{J}^{\delta\beta}{}_{\delta}$
Total # constraint(s):	13	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{24\overset{(4)}{\kappa}_1^2 + 8\overset{(4)}{\kappa}_1 (2\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) + (2\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2}{\overset{(4)}{\kappa}_1 (6\overset{(4)}{\kappa}_1 (\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4) + (2\overset{(4)}{\kappa}_3 + \overset{(4)}{\kappa}_4)^2)}$

Resolved unitarity condition(s):

$$(\overset{(4)}{\kappa}_4 < 0 \;\&\&\; ((\overset{(4)}{\kappa}_3 < -\frac{\overset{(4)}{\kappa}_4}{2} \;\&\&\; 0 < \overset{(4)}{\kappa}_1 < \frac{-4\overset{(4)}{\kappa}_3^2 - 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4^2}{6\overset{(4)}{\kappa}_3 + 6\overset{(4)}{\kappa}_4}) \;||\; (-\frac{\overset{(4)}{\kappa}_4}{2} < \overset{(4)}{\kappa}_3 < -\overset{(4)}{\kappa}_4 \;\&\&\; 0 < \overset{(4)}{\kappa}_1 < \frac{-4\overset{(4)}{\kappa}_3^2 - 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4^2}{6\overset{(4)}{\kappa}_3 + 6\overset{(4)}{\kappa}_4}) \;||$$
$$(\overset{(4)}{\kappa}_3 == -\overset{(4)}{\kappa}_4 \;\&\&\; \overset{(4)}{\kappa}_1 > 0) \;||\; (\overset{(4)}{\kappa}_3 > -\overset{(4)}{\kappa}_4 \;\&\&\; (\overset{(4)}{\kappa}_1 < \frac{-4\overset{(4)}{\kappa}_3^2 - 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4^2}{6\overset{(4)}{\kappa}_3 + 6\overset{(4)}{\kappa}_4} \;||\; \overset{(4)}{\kappa}_1 > 0)))) \;||$$
$$(\overset{(4)}{\kappa}_4 == 0 \;\&\&\; ((\overset{(4)}{\kappa}_3 < 0 \;\&\&\; 0 < \overset{(4)}{\kappa}_1 < -\frac{2\overset{(4)}{\kappa}_3}{3}) \;||\; (\overset{(4)}{\kappa}_3 > 0 \;\&\&\; (\overset{(4)}{\kappa}_1 < -\frac{2\overset{(4)}{\kappa}_3}{3} \;||\; \overset{(4)}{\kappa}_1 > 0)))) \;||$$
$$(\overset{(4)}{\kappa}_4 > 0 \;\&\&\; ((\overset{(4)}{\kappa}_3 < -\overset{(4)}{\kappa}_4 \;\&\&\; 0 < \overset{(4)}{\kappa}_1 < \frac{-4\overset{(4)}{\kappa}_3^2 - 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4^2}{6\overset{(4)}{\kappa}_3 + 6\overset{(4)}{\kappa}_4}) \;||$$
$$(\overset{(4)}{\kappa}_3 == -\overset{(4)}{\kappa}_4 \;\&\&\; \overset{(4)}{\kappa}_1 > 0) \;||\; (-\overset{(4)}{\kappa}_4 < \overset{(4)}{\kappa}_3 < -\frac{\overset{(4)}{\kappa}_4}{2} \;\&\&\; (\overset{(4)}{\kappa}_1 < \frac{-4\overset{(4)}{\kappa}_3^2 - 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4^2}{6\overset{(4)}{\kappa}_3 + 6\overset{(4)}{\kappa}_4} \;||\; \overset{(4)}{\kappa}_1 > 0)) \;||$$
$$(\overset{(4)}{\kappa}_3 == -\frac{\overset{(4)}{\kappa}_4}{2} \;\&\&\; (\overset{(4)}{\kappa}_1 < 0 \;||\; \overset{(4)}{\kappa}_1 > 0)) \;||\; (\overset{(4)}{\kappa}_3 > -\frac{\overset{(4)}{\kappa}_4}{2} \;\&\&\; (\overset{(4)}{\kappa}_1 < \frac{-4\overset{(4)}{\kappa}_3^2 - 4\overset{(4)}{\kappa}_3\overset{(4)}{\kappa}_4 - \overset{(4)}{\kappa}_4^2}{6\overset{(4)}{\kappa}_3 + 6\overset{(4)}{\kappa}_4} \;||\; \overset{(4)}{\kappa}_1 > 0))))$$